

K CHAITRA LATHA

Data Scientist · ML Engineer · Python & Backend Developer

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PROFESSIONAL SUMMARY

Results-driven Data Scientist and Software Developer with hands-on experience building scalable ML systems, REST APIs, analytics dashboards, and enterprise automation platforms. Proficient in Python, TensorFlow, Scikit-learn, SQL, and Apache Kafka, with a strong track record of delivering end-to-end data pipelines, workflow automation, and production-grade backend services. Adept at collaborating cross-functionally to translate complex business requirements into reliable, data-driven solutions.

PROFESSIONAL EXPERIENCE

Associate Software Developer | ATOM

Mar 2025 – Present

Promoted to full-time role from August 2025, following internship (Mar 2025 – Jul 2025)

- ▶ Engineered end-to-end features for three enterprise platforms (LMS, AMS, QMS) — spanning requirement analysis, development, testing, and production deployment.
- ▶ Delivered the complete candidate interview invitation flow for INTERVEXA AI: built an auth-aware React validation page with secure backend API verification, conditional routing for unauthenticated users, and polished async UX (loading states, error handling).
- ▶ Designed and deployed Python-based REST APIs supporting automation, analytics, exam scheduling, and data enrichment workflows — improving operational throughput across teams.
- ▶ Architected ITSM-aligned workflow automation (incident handling, multi-level approvals, real-time notifications, escalation rules) using Flow Designer patterns — reducing manual process overhead.
- ▶ Developed analytics dashboards and structured performance reports for AMS and LMS; visualized KPIs enabling data-driven stakeholder decisions.
- ▶ Built real-time data pipelines for exams, evaluations, and user activity — supporting live monitoring and audit-ready reporting.
- ▶ Implemented Role-Based Access Control (RBAC) across systems, enforcing secure, permission-based data access for 3+ platform modules.
- ▶ Resolved critical database integration issues across PostgreSQL and MySQL environments, improving data consistency and system uptime.
- ▶ Improved frontend responsiveness across HTML/CSS/JavaScript UIs through close collaboration with product teams, accelerating feature delivery cycles.
- ▶ Produced technical documentation and led walkthroughs for workflow automation features, reducing onboarding time for new stakeholders.

Technical Intern | ATOM

Mar 2025 – Jun 2025

- ▶ Assisted senior developers in backend development, debugging, and feature enhancements across LMS and AMS production systems using Python and Java.
- ▶ Contributed to REST API development and backend service integration as part of the core engineering team.
- ▶ Gained hands-on experience with production deployments, code reviews, and sprint-based development cycles.
- ▶ Supported data pipeline design and analytics reporting tasks, building foundational skills in data engineering and system integration.

EDUCATION

Master of Computer Applications (MCA) — Data Science

2023 – 2025

Chanakya University, Bangalore · CGPA: 9.13 / 10

Relevant Coursework: Machine Learning, Deep Learning, Statistical Modelling, real time Analytics, Database Systems

Bachelor of Computer Applications (BCA)

2020 – 2023

Shree Medha Degree College, Ballari — VSK University · CGPA: 9.24 / 10

Relevant Coursework: Data Structures, Algorithms, Object-Oriented Programming, Web Technologies, DBMS

TECHNICAL SKILLS

- **Languages:** Python, Java, C, C++, C#.NET
- **ML / AI:** Scikit-learn, TensorFlow, Keras, Neural Networks, CNN, K-Means, KNN, Feature Engineering
- **Data Analysis:** Pandas, NumPy, SQL, Statistical Modeling, Time-Series Analysis
- **Visualization:** Tableau, Power BI, Apache Superset, Matplotlib
- **Big Data:** PySpark, Apache Kafka
- **Databases:** PostgreSQL, MySQL
- **Web / APIs:** Flask, REST APIs, React, HTML, CSS, JavaScript, TypeScript, Workflow Automation
- **Tools & Platforms:** Git, Jupyter Notebook, VS Code, Ubuntu
- **Soft Skills:** Cross-functional collaboration, technical documentation, stakeholder communication, mentoring

KEY PROJECTS

- ▶ **Financial Personality Segmentation:** Clustered customer spending behavior using K-Means and feature engineering, enabling intelligent financial profiling and targeted product recommendations.
- ▶ **Stock Market Direction Prediction:** Built an ML classification model to predict price movement direction using technical indicators and time-series analysis with Scikit-learn; achieved consistent directional accuracy on historical data.
- ▶ **CNN-Based Digit Recognition:** Designed and trained a convolutional neural network on the MNIST dataset (TensorFlow/Keras) with image preprocessing; achieved high classification accuracy on handwritten digits.
- ▶ **Brain Tumor Risk Analysis:** Applied statistical ML techniques to predict brain tumor risk factors, supporting early clinical decision-making through probabilistic risk scoring.
- ▶ **COVID-19 Outcome Prediction:** Conducted comparative analysis of KNN variants (Euclidean, Manhattan, Minkowski) for patient outcome classification; identified optimal distance metrics for clinical datasets.
- ▶ **Stock & Weather Analytics Pipeline:** Built a real-time data ingestion and analytics pipeline using Apache Kafka APIs for concurrent stock and weather data processing.
- ▶ **Geospatial ML Analysis:** Classified satellite imagery using ML algorithms to identify land-use patterns, applying preprocessing and feature extraction techniques for spatial datasets.
- ▶ **Service Management System:** Developed an end-to-end incident management and service catalog platform aligned with ServiceNow ITSM modules, supporting full ticket lifecycle management.
- ▶ **E-Vote Cast:** Designed a secure Java-based online voting system with real-time admin controls, vote integrity verification, and comprehensive audit logging.

RESEARCH WORK

- ▶ **KNN for COVID-19 Outcome Prediction:** Comparative study of K-Nearest Neighbor algorithm variants using Python and Scikit-learn; evaluated distance metrics and normalization strategies for clinical classification tasks.
- ▶ **Digit Classification using Neural Networks:** Implemented handwritten digit classification on MNIST using a Multi-Layer Perceptron with ReLU activation and backpropagation; analyzed the effect of depth and regularization on accuracy.

CERTIFICATIONS

- ▶ **Advanced Content-Based Recommender System Essentials** — LetsUpgrade (NSDC & GDG MAD) | Feb 2026
- ▶ **Scikit-learn Classification Essentials: Imbalanced Data & Pipelines** — LetsUpgrade (NSDC & GDG MAD) | Feb 2026
- ▶ **Python ATM Simulation Essentials Mini Project** — LetsUpgrade (NSDC & GDG MAD) | Feb 2026
- ▶ **Data Science with Python** — LetsUpgrade (NSDC & GDG MAD) | Jun 2025

AWARDS & ACHIEVEMENTS

- ▶ Head Coordinator for Bitotsav 2k23 — led a state-level technical fest, managing cross-team logistics and ensuring successful execution of all technical events.
- ▶ Certificate of Appreciation from NxtWave for leadership and successful execution of an AI workshop attended by 100+ participants.
- ▶ Core member of Team aTsom for the MIT Hackathon initiative — contributed end-to-end from ideation to final presentation.

- Formal appreciation from Director of Learning & Development for technical excellence, teamwork, and measurable platform improvements.
- Mentored a student team through the Amaravati Quantum Valley Hackathon pre-selection stage, guiding project scoping and technical strategy.